



House Price Prediction - Task 2

Feature Engineering, Model Optimization & Performance Comparison

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

In [3]:

```
data = fetch_california_housing(as_frame=True)
df = pd.concat([data.data, data.target.rename("HousePrice")], axis=1)

print(df.head())
print("Shape:", df.shape)
print("Columns:", df.columns)
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude \
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85

	Longitude	HousePrice
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

Shape: (20640, 9)

Columns: Index(['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude', 'HousePrice'], dtype='str')

In [4]:

```
X = df.drop("HousePrice", axis=1)
y = df["HousePrice"]
```

In [5]:

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [6]:

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

In [7]:

```
models = {
    "Linear Regression": LinearRegression(),
```

```

    "Ridge Regression": Ridge(alpha=1.0),
    "Decision Tree": DecisionTreeRegressor(max_depth=5)
}

```

In [9]:

```

results = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    predictions = model.predict(X_test)

    # Compute metrics
    mse = mean_squared_error(y_test, predictions)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, predictions)

    results[name] = {"RMSE": rmse, "R2 Score": r2}

results_df = pd.DataFrame(results).T
print(results_df)

```

	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

In [10]:

```

best_model = LinearRegression()
best_model.fit(X_train, y_train)
y_pred = best_model.predict(X_test)

plt.figure(figsize=(6,6))
plt.scatter(y_test, y_pred, alpha=0.4)
plt.xlabel("Actual House Prices")
plt.ylabel("Predicted House Prices")
plt.title("Actual vs Predicted House Prices")
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color="red")
plt.show()

```



In []: